

SRI SAI SNIGDHA ANANTHARAJU

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PROFESSIONAL SUMMARY

I'm an urban data scientist who uses geospatial analysis, machine learning, and civic tech to tackle complex challenges in cities. I bring a consulting mindset while translating messy urban data into insights for planners, policymakers, and public agencies to drive inclusive, data-informed change. Proven leadership as President of the ASPRS NYU Student Chapter and the Applied Urban Science Association (AUSA) at NYU Tandon.

EDUCATION

New York University, Tandon School of Engineering

New York, NY

Master of Science (MS), Urban Data Science [GPA: 3.98/4]

May 2026

Relevant Coursework: Urban Spatial Analytics, Advanced Spatial Analytics, Machine Learning for Cities, Civic Analytics

- Recipient of CUSP Merit Scholarship of \$18,000
- Winner, *Best Technical Contribution*, Global Data Dive at King's College London
- Recipient, CUSP Experiential Scholarship for transportation research under Prof. Anton Rozhkov
- Winner, GIS Mapping Competition, New York University

School of Planning and Architecture, Vijayawada

Vijayawada, India

Bachelor of Planning, Urban and Regional Planning [GPA: 3.88/4]

May 2024

Relevant Coursework: Advanced Geoinformatics and Studio

RELEVANT EXPERIENCE

NYMTC 9/11 Memorial Transportation Research Fellow, NYC Department of Transportation

October 2025-present

- Conduct citywide crash analysis, roadway geometry review, high-injury network assessments and vehicular and pedestrian data processing to support Vision Zero decision-making.
- Perform GIS- and remote-sensing based existing conditions analysis, integrating streetscape imagery, and field observations to identify pedestrian and vehicular risk factors.
- Apply photogrammetry workflows to measure sight-distance constraints, turning radii, lane widths, vertical/horizontal clearance, and intersection visibility.
- Produce technical maps, visualizations and reports for DOT teams to support scoping, planning, and safety project development.

Road Safety Researcher, Center for Urban Science + Progress, New York University

September 2025 – present

- Conduct analysis of intersections using photogrammetric techniques to map pedestrian-vehicle interactions.
- Generate 3D reconstructions of roadway environments from drone imagery and street-level data to create realistic models for simulation.
- Integrate GIS-derived datasets (road geometry, traffic density, pedestrian flows) into VR simulations to evaluate safety interventions.
- Apply spatial metrics to identify high-risk zones and patterns of conflict, informing urban safety planning.
- Present findings through maps, 3D visualizations, and technical reports, demonstrating the application of GIS and photogrammetry to transportation research.

GIS Consultant, Data Services, New York University

May 2025-present

- Deliver technical GIS support and spatial data consultation across research projects.
- Facilitate spatial data processing analysis, design, and using ESRI Products, QGIS and open-source tools.
- Lead workshops and create instructional materials on GIS methodologies, tools, and applications to advance spatial literacy and data-driven research.

Graduate Assistant, Center for Urban Science + Progress, New York University

November 2024 – present

- Organized and categorized 100+ spatial datasets for reuse in climate, infrastructure, and mobility research.
- Coordinate logistics for events and programs, ensuring seamless execution while managing communications.

Data Science & Machine Learning Intern, School of Planning and Architecture, Bhopal

June 2023 – July 2023

- Extracted and analyzed 5,000+ accessibility-related data points using Python (BeautifulSoup, Selenium).
- Applied BERT, VADER, and GAN models to classify and evaluate heritage accessibility compliance.
- Supported the development of a Government of India accessibility mapping tool impacting 10,000+ users.

- Automated accessibility assessments with GANs and R-CNN, boosting analysis efficiency by 30%.

Urban Planning Intern, City Planning Department, Vijayawada Municipal Corporation *June 2022 – August 2022*

- Conducted traffic flow analysis & proposed vending zones, NMT lanes leading to 12% reduction in traffic accidents.
- Contributed to the corridor improvement plan by integrating pedestrian and non-motorized transport-friendly features leading to around 25% increase in pedestrian safety and smoother traffic flow along 6+ critical corridors.
- Facilitated 4+ cross-sectoral stakeholder meetings to develop a TOD plan, impacting 25,000+ daily commuters.

SKILLS

- **Programming & GIS:** Python, ESRI Suite, QGIS, Google Earth Engine, GeoDa, EPANET, R Studio
- **Other technical skills:** SQL, Tableau, Power BI, MS Office, Adobe Suite, Alteryx, AnyLogic, PTV Visum, AutoCAD
- **Soft Skills & Communication:** Public Speaking, Workshop Facilitation, Interdisciplinary Collaboration

RELEVANT PROJECTS

Forensic Geosimulation: Finding the Factual Boundary of Road Accidents *December 2025*

- Engineered a hybrid spatial-kinematic pipeline to bound witness claims by physical possibility, combining Line-of-Sight (LOS) analysis with Monte Carlo simulations.
- Integrated multi-source urban data including NYC PLUTO (building footprints as static occluders) and TIGER street centerlines to model dynamic driver approach paths and 3D visibility.
- Simulated 2,000 scenarios using Python (NumPy/Matplotlib) to evaluate the impact of heterogeneous human behaviors based on AASHTO standards.

SoftWhere: Optimizing Housing Decisions to Minimize Commute Times *May 2025*

- Designed a geospatial decision-support tool recommending housing locations based on subway commute efficiency using GTFS data and reverse-time Dijkstra algorithm.
- Modeled NYC's subway network as a time-dependent graph, integrating MTA schedules & spatial proximity metrics.
- Developed commute accessibility scores for over 2,000 hexagons, visualized using interactive heatmaps to identify transit-rich yet underappreciated neighborhoods
- Proposed planning applications for housing equity, sustainable commuting, and transit-oriented location strategies.

Invisible Boundaries: Network Analysis of Healthcare Access in Houston, TX *March 2025*

- Developed a network model in ArcGIS Pro to analyze healthcare accessibility in Houston, TX.
- Computed least-time travel paths to healthcare facilities, incorporating various speed limits and road hierarchy.
- Performed spatial inequity assessments using service area analyses, pinpointing underserved regions with >20-minute emergency response delays, highlighting demographic and spatial disparities.

New York: Skyline vs. Sustainability – Factors affecting NYC Buildings' Energy Efficiency *February 2024*

Won 'Best Technical Contribution' at the Global Data Dive (3-day Hackathon) held by King's College London

- Preprocessed building energy data (LL84) for 10000+ buildings in Manhattan, NYC
- Developed a hybrid deep learning model combining Convolutional Neural Networks (CNN) for spatial features and Multi-Layer Perceptrons (MLP) for intrinsic building attributes to predict building energy efficiency.
- Analyzed the impact of intrinsic and external factors on building energy efficiency in New York City.
- Assessed Local Law 97 compliance, identifying \$218M in annual energy costs and associated penalties.
- Developed data-driven insights to help policymakers, building owners, and investors make decisions on energy.

Vijayawada City-Wide Freight Transport Analysis and Emission Mitigation Strategies *May 2024*

- Mapped spatial distribution of pollutants using ArcGIS, pinpointing high-emission zones.
- Optimized logistics by evaluating trip patterns of 65 daily freight vehicles, with medium commercial vehicles averaging 1,125 km per trip, and proposed solutions to reduce congestion and improve routing efficiency.
- Analyzed freight operations across Vijayawada, calculating vehicle emissions of 9,118 kg/day, including PM10, PM2.5, CO, NOx, and SO2, and developed phased EV adoption plans to achieve a 62% reduction in emissions

PUBLICATION

Saha, K., Anantharaju, S.S.S., Khare, R. (2025). GIS to Machine Learning: Application of Emerging Tools to Assess the Accessible Built Environment in Indian Cities. In: Yadav, V., Mishra, M., Gavske, K.K. (eds) The Built Environment and Public Health in India. New Frontiers in Regional Science: Asian Perspectives, vol 85. Springer, Singapore.

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